Data profiling before integration

DAT 325 Project Two | Academic coursework | Profiling findings and correction plan

An academic review of data from an acquired-firm scenario, intended to identify quality problems before integration. The report documents Python and pandas profiling and proposes source-verified corrections.

Completed analysis documented

Reviewed Age, Marital_Status, Days_Employed and Citizen_Desc for missing values, invalid categories and unusual numeric values. The report names pandas missing-value and descriptive-statistics methods as part of the profiling approach.

Reported findings

- Age: a missing value, a negative value of -39 and an extreme value of 139.
- Marital_Status: four missing values and the invalid category Female.
- Days_Employed: a missing value, a negative value of -514 and an extreme value of 52,246.
- Citizen_Desc: a missing value and the invalid category Male.

Correction decisions

Verify affected entries against reliable source records. Correct only values that can be confirmed, document unresolved records, and investigate extreme values before deciding whether to change or remove them. The report does not support automatic deletion or guessed replacement values.

Proposed transformations

Split Employee_Name into separate first-name and last-name fields, trim extra spaces, and standardize DOB to the destination date format before a future merge. These transformations were specified, not demonstrated as completed.

Scope and evidence

The evidence is the written profiling report. No notebook, Python script, input dataset, execution log or validated output was available to reproduce the results. The report does not establish a completed cleanup or successful merge. No personal employee records are included here.

Skills demonstrated

Data profiling; anomaly assessment; validation planning; transformation mapping.

Source and attribution

Olga Solodovnikova, Project Two Executive Summary Report, DAT 325, Southern New Hampshire University, October 2, 2026. The report cites pandas DataFrame.isnull and DataFrame.describe documentation.